

MAE 280B

Linear Control Design – Winter 2013

Final

Instructions:

- Due on 03/20/2013 at my office (EBU I 1602) by 5:00 PM;
- Use Matlab or Mathematica;
- You get marks for clarity;
- You loose marks for obscurantism;
- This exam has 16 questions, for a total of 38 points and 5 bonus points.

Part I

Delay-System. Consider the linear delay-system

$$\dot{x}(t) = Fx(t) + Gx(t - \tau), \quad \tau > 0 \quad x(t) = \phi(t), \quad t \in [-\tau, 0], \quad \phi(t) \not\equiv 0$$

and answer the following questions:

1. (1 point (bonus)) What is the *state* of the delay-system? What is the dimension of the state?
2. (2 points (bonus)) Let

$$V_\tau(x(t)) = \int_{t-\tau}^t x(\theta)^T Q x(\theta) d\theta.$$

Show that

$$\dot{V}_\tau(x(t)) = x(t)^T Q x(t) - x(t - \tau)^T Q x(t - \tau).$$

3. (2 points) Let

$$V(x(t)) = x(t)^T P x(t) + V_\tau(x(t)).$$

Show that

$$\dot{V}(x(t)) = x(t)^T (F^T P + P F + Q) x(t) + 2x(t)^T P G x(t - \tau) - x(t - \tau)^T Q x(t - \tau).$$

4. (2 points) **Delay-Independent Stability.** Show that if there exists $P \succ 0$, and Q such that

$$\begin{bmatrix} F^T P + P F + Q & P G \\ G^T P & -Q \end{bmatrix} \prec 0 \quad (1)$$

then $V(x(t)) > 0$ and $\dot{V}(x(t)) < 0$ for all $x(t) \not\equiv 0$. That is, the origin is an asymptotically stable equilibrium point for the delay-system for any $\tau \geq 0$, or simply, the delay-system is asymptotically stable for any $\tau \geq 0$.

5. (2 points) Show that if there exists $P \succ 0$ and Q such that LMI (1) is true then F is Hurwitz.
6. (4 points) Consider the feedback connection of a linear system with output y and input u and transfer-function

$$H(s) = \frac{1}{s^2 + 2\zeta s + 1}$$

with the scalar controller

$$u(t) = -k \dot{y}(t - \tau)$$

Represent the feedback connection as a linear delay-system. Solve the LMI (1) to determine what is the maximum value of k for which the closed-loop system is stable independently of the delay for $\zeta = \{\sqrt{2}/2, 1, 2, -1/2\}$. Comment your findings.

7. (2 points (bonus)) Repeat Question 6 using Bode plots or the Nyquist criterion.
8. (2 points) **Delay-Independent Stabilization.** Consider the LTI system

$$\dot{x}(t) = Ax(t) + Bu(t), \quad \tau > 0 \quad x(t) = \phi(t), \quad t \in [-\tau, 0], \quad \phi(t) \neq 0,$$

and the delayed state-feedback controller

$$u(t) = Kx(t - \tau), \quad \tau > 0.$$

Show that if A is not Hurwitz, then it is not possible to find K such that the closed-loop system is proved stable by the delay-independent LMI (1).

9. (2 points) **Delay-Dependent Stability.** It is known (this one here is a variation of a result in Gu, Kharitonov & Chen 2003) that if there exists $P \succ 0$, Q , W and $r \geq 0$ such that

$$\begin{bmatrix} F^T P + PF + Q - W & PG + W & rF^T W \\ G^T P + W & -Q - W & rG^T W \\ rWF & rWG & -W \end{bmatrix} \prec 0, \quad \begin{bmatrix} Q & rG^T W \\ rWG & W \end{bmatrix} \succ 0 \quad (2)$$

then the origin is an asymptotically stable equilibrium point for the delay-system for all $0 \leq \tau \leq r$, i.e. the delay-system is asymptotically stable for any $0 \leq \tau \leq r$.

Show that if there exists $P \succ 0$ and Q such that the LMI (1) is true then there exists $P \succ 0$, Q , Z , R and $r > 0$ such that (2) is also true. Interpret the result.

10. (4 points) Consider the feedback connection described in Question 6. Set $k = 2$ and solve the inequality (2) and determine the maximum value of $r > 0$ for which it can prove that the closed-loop system is stable for $\zeta = \{\sqrt{2}/2, 1, 2, -1/2\}$. Comment your findings.

Part II

A platoon of vehicles on a freeway is modeled as:

$$m_i \dot{v}_i(t) = u_i(t) + w_i(t), \quad i = 1, \dots, N,$$

where v_i , m_i , u_i , and w_i is the i th vehicle's velocity, mass, control input, and perturbation. You want to determine whether a control law of the form

$$u_1 = -k_1(v_1 - \bar{v}), \quad u_i = -k_i(v_i - v_{i-1}), \quad i > 1, \quad (3)$$

stabilizes the platoon and is optimal in some way.

1. (3 points) Compute a state-space model for the platoon where the state vector is

$$x = \begin{pmatrix} v_1 - \bar{v} \\ v_2 - v_1 \\ v_3 - v_2 \\ \vdots \\ v_N - v_{N-1} \end{pmatrix}.$$

2. (4 points) Using what you know about decentralized control, show how the control (3) can be interpreted as a decentralized control. Assuming that the perturbations $w_i(t)$ are white noise sources with unit covariance matrix, formulate two distinct LMIs that can be used to compute the gains k_i 's while minimizing an upper bound to the cost function:

$$J = \lim_{t \rightarrow \infty} E\{x(t)^T x(t) + u(t)^T u(t)\}.$$

Interpret the two components of the cost function in the case of the platoon. Interpret the constraint you had to impose on the Lyapunov variable X .

3. (3 points) Solve the LMIs you formulated in Question 2 when $m_i = m = 1$ are the same for all vehicles and $N = 4$. Explain the results you obtained.
4. (3 points) Solve the LMIs you formulated in Question 2 when $m_i = i$ and $N = 4$. Explain the results you obtained.
5. (3 points) Solve the LMIs you formulated in Question 2 when $m_i = N - (i - 1)$ and $N = 4$. Explain the results you obtained.
6. (4 points) What problem have you solved? What does this control law guarantees in terms of the spacing between the vehicles? How would you modify the problem so that the space between the vehicles could also be controlled? You do not need to solve the problem, but you need to provide evidence that it can be worked out.